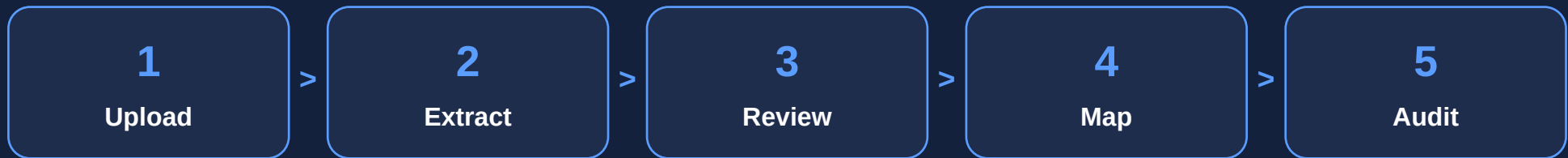


PROTOTYPE

Land Fraud: India's Hidden Crisis

A forensic verification platform to digitize, detect and prove land ownership in Tamil Nadu.
Screen-by-screen prototype of the pipeline described in the presentation.



Problem: 15-45 days of manual verification, handwritten registers, no citizen self-service.

Solution: OCR + confidence scores, human review under 85%, GeoJSON map with overlap detection, hash-chained audit trail.

Planned stack: PostgreSQL + PostGIS, Leaflet/MapLibre, public API for Bhoomi / DILRMP / Bhulekh.

Upload

The citizen or officer drops a scanned Tamil Nadu register page into the portal. No preprocessing needed.

- Accepts raw scans (JPG, PNG, PDF)
- Automatic deskew and denoise in the preprocess stage
- Page is queued for OCR immediately

Portal / New verification

Drop scanned register page here

or click to browse

Village: Kancheepuram Taluk: Sriperumbudur

Pattadar: Murugan Selvam S.No. 142/3B

Extent: 1.2_ ha (ink smudged) Reg. 14-03-1998

Boundaries: N - S.No. 142/2, S - Road, E - 143

Start OCR

OCR Extraction

Fields appear one by one with a confidence score, producing a structured JSON record per page.

- Owner name (Tamil and transliterated), survey number, area, boundaries, registration date
- Survey number format validated
- Fields below 85% are flagged automatically

Extraction in progress

Processing page 1 ... 5 of 5 fields extracted

Owner name	Murugan Selvam		96%
Survey no.	142/3B		91%
Area (ha)	1.24		78%
Boundaries	N:142/2 S:Road E:143		82%
Reg. date	14-03-1998		97%

```
{ "survey_no": "142/3B", "area_ha": 1.24, "confidence_min": 78 }  
{ "flagged": ["area_ha", "boundaries"] }
```

Validation & Human Review

Low-confidence fields light up in red. A reviewer corrects them before anything is saved.

- Cross-field checks and duplicate survey IDs flagged instantly
- Queue prioritised by dispute risk score
- Confirmed records are written to PostgreSQL + PostGIS

Reviewer dashboard

2 fields need review (confidence below 85%)

Owner name

Murugan Selvam

96%

Survey no.

142/3B

91%

Area (ha)

1.24

78%

Boundaries

N:142/2 S:Road E:143

82%

Reg. date

14-03-1998

97%

Confirm & save record

Edit

Map & Overlap Detection

The verified parcel appears as a polygon.
Conflicting claims are highlighted in real time.

- GeoJSON polygons on a Leaflet/MapLibre map (schematic here)
- Overlap detection replaces manual cross-referencing
- Citizen portal shows ownership with tamper evidence



Immutable Audit Trail

Every verification event is chained to the previous one: tamper-evident, time-stamped and legally defensible.

- Editing any entry breaks every later hash
- Public API exposes verification status to Bhoomi, DILRMP and Bhulekh
- No rip-and-replace of existing systems

Audit log

#1 Page uploaded

prev 0000...0000

hash a41f9c3e

#2 OCR complete

prev a41f9c3e

hash 7be20d15

#3 Reviewer confirmed S.No. 142/3B

prev 7be20d15

hash c93a6f08

#4 Parcel published, verified badge

prev c93a6f08

hash 1d58e2b7

#5 Overlap flagged with S.No. 143

prev 1d58e2b7

hash f0276a9c